Cover letter

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To the Editor-in-Chief, *Science and Medicine in Football*

Dear Editor,

I submit the Original Investigation **"What can be measured from the broadcast? Validity of single-camera video for football-kick kinematics and swing-leg joint loads"** for consideration in *Science and Medicine in Football*.

Clubs, leagues and technology vendors increasingly derive technique and load metrics from broadcast video, and since the 2026/27 season LaLiga timestamps every touch with a connected ball. Whether the kicking body can be measured from the same broadcast has not been tested against a reference. This method-comparison study asks a practical question for football: for which uses — comparison with laboratory norms, ranking players or monitoring an individual — are kicking kinematics and swing-leg joint loads estimated from a single broadcast-like camera acceptable, and how large would a study validating a broadcast-derived load score as an injury predictor have to be?

The main findings are:

- From a single broadcast camera, none of six kinematic features or three swing-leg joint moments met exploratory operational criteria informed by the literature for any of the three uses. The detector error on 200 Premier League shots was 7.6–12.9 % of player height.
- Three simulated replays with laboratory correction met selected exploratory criteria for between-participant ranking of knee flexion and laboratory comparison of trunk leans. Reference contact timing reduced error by about a quarter relative to reconstructed automatic timing; no connected-ball sensor or synchronisation was evaluated. The tested 25-versus-50-fps contrast did not improve error in this experiment.
- The laboratory KLI association with ankle speed ($\rho = 0.79$) is a within-model consistency check with shared kinematic input. Broadcast agreement was poor (ICC 0.090; calibration slope $b = 0.081$). These slopes do not estimate reliability; the earlier KLI-specific event counts and league-season extrapolations are withdrawn. Only independent hypothetical information fractions are used for event-count sensitivity; no injury outcomes or injury-risk validity were assessed.

Reference data comprise seven CMU ball kicks from two people and 83 MoVi air kicks. Real-video validation uses 162 recordings from 82 MoVi participants, with a per-recording digital twin (aggregate error association $r = 0.84$). Agreement is reported following GRRAS, with equivalence tests and a custom paired sign-flip waveform permutation procedure, not the spm1d package. Scale, alignment, kicking leg and known contact use reference information in the main conditions. Thirteen smartphone recordings of the author's own kicks illustrate the pipeline without a reference measurement. The criteria are exploratory and do not establish individual clinical, sporting or injury-risk validity. Code and the broadcast residual bank are openly available (https://github.com/convderover/broadcast-kick-biomechanics; Zenodo DOI 10.5281/zenodo.23439408); the statistical outputs and GRRAS checklist accompany the repository.

A preprint was published on SportRxiv on 26 October 2026 (DOI 10.51224/SportRxiv.1131); it has not been peer reviewed. This is an unsent cover-letter draft, not evidence of submission or acceptance. The study is a secondary analysis of public datasets and was not pre-registered. The extensive use of generative AI (study design proposals, code, analyses, figures and drafting) is declared in detail in the manuscript, and all code and outputs are public.

The manuscript is original, has not been published and is not under consideration elsewhere. I am its sole author, I report no competing interests and the work received no funding.

Thank you for considering this work.

Yours sincerely,

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